Control is an illusion.

Confidence without control: scientific background and industry position.

← all documents · Kat Titterton · San Francisco

ABSTRACT

Every result in cell biology is a difference from a control, and nothing checks the control against anyone else's. On the public JUMP Cell Painting dataset, a classifier names the lab from untreated wells alone 99.9% of the time against 9.1% chance, so "normal" is a lab-specific coordinate, not a shared one. The normalisations that make cross-lab discovery possible erase that coordinate (centring each plate on its own controls drops lab identity to 24%): right for discovery, wrong for quality control. Datum Line keeps the coordinate and uses it. A lab's control-well images are placed against the published field and every lab running the same line, and the lab gets back a certificate, a distance, and the direction of any shift. The bet still to be proven is that label-free brightfield carries the same signal as five-stain fluorescence; the test is specified and pending.

1 The control problem

A control removes what a plate shares, including the plate's own state, so a sick plate hides behind its own controls. Write a measured well as a shared mean, a plate state, and a treatment effect that may depend on that state:

$$x_{test} = \mu + s + \tau(s) \qquad x_{ctrl} = \mu + s \qquad \Delta = x_{test} - x_{ctrl} = \tau(s) \tag{1}$$

The state s cancels from the number but not from the cells. If s is off, Δ is still computed, still looks clean, and is measured on cells that are not the ones the field has in mind.

The same logic nests upward: well inside plate, plate inside batch, batch inside site, site inside the field. Each control sees only inside its own box. A whole plate off is visible to the batch; a whole site off is visible to nobody in the building. Only a reference outside every box, the field, can see where a lab sits.

This is not an exotic failure. Irreproducible preclinical research costs about $28B a year in the US alone, and the largest single contributor, about 36%, is biological reagents and reference materials [1]. Cells are the input nobody certifies against anyone else's.

2 Morphological profiling as a sensor

An image of a cell is an unbiased, many-dimensional sensor: it answers "is anything about this run off?" rather than one hand-picked question. Cell Painting multiplexes six fluorescent dyes in five channels to mark eight cellular components, and automated analysis measures about 1,500 features per cell: size, shape, texture, intensity [2].

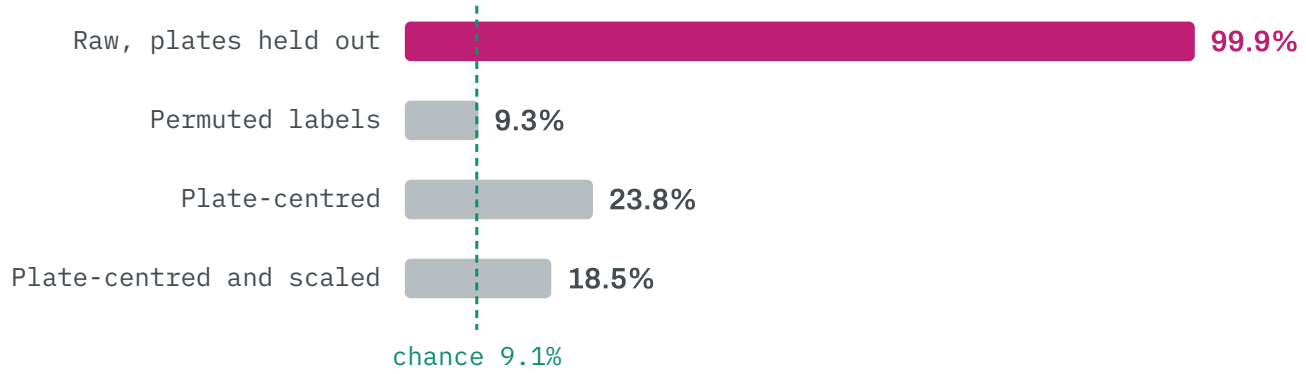
At scale, the reference already exists. The JUMP Cell Painting Consortium (ten pharmaceutical companies, six technology partners, two non-profits) profiled over 116,750 compounds plus gene knockouts and overexpressions, all in U2OS cells [3]. Its main compound set, cpg0016, was produced across 11 sites, which is exactly the variation a cross-lab reference needs, and it is public under CC0 [3].

The sensor is not the bottleneck. Many widely used bioactivity benchmarks are predicted well by simply counting cells, because their assays track cell health [5]. When a naive baseline matches a model, the comparison is the problem, not the model. In the author's hepatocyte work, three feature extractors (CellProfiler, a CNN, a ViT) performed similarly [6]. The asset is the reference, not the extractor.

3 Evidence on public data

Untreated wells carry their lab's fingerprint, and the normalizations built for discovery erase it.

Fig. 1 **Lab identity from untreated wells, before and after correction**

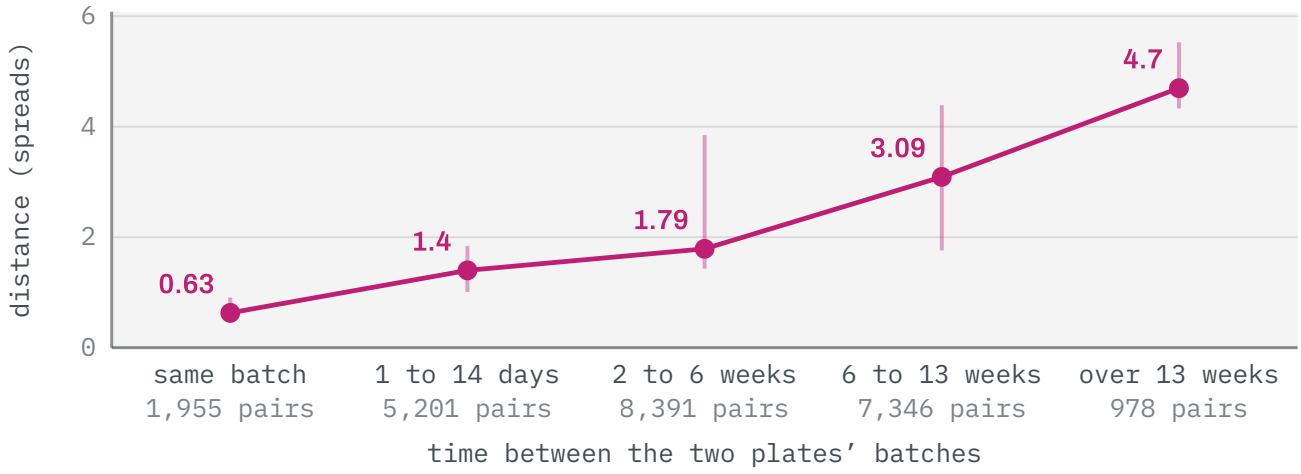


Balanced accuracy naming one of 11 labs from untreated JUMP wells; 93,228 wells, 3,773 features, plates held out. Normalisation and permutation tests on 29,943 wells. results/e1, e1-redteam.

LAB 2, OCTOBER 2021 BATCH	RESULT	FILE
Cells per untreated well	182; inside the lab's range, no local QC rule fires	results/e2-timeline
Distance from the lab's own June baseline	8.1 within-plate spreads	results/e2-timeline
Distance to the nearest other lab (lab 8)	5.1	results/e2-timeline
Wells still named as their own lab	100%	results/e1

Space. The batch above passes the check its own lab runs, yet sits further from the lab's own June baseline than from another lab's centre. Local QC cannot see this by construction (section 1).

Fig. 2 **Even one lab drifts**



Median distance between plate centroids of one lab, in within-plate spreads, with the interquartile range; lab 2, 219 plates, 13 batches, June to October 2021. results/e1b-drift.

Time. Within one lab, the distance between two plates grows from 0.63 in the same batch to 3.09 at 6 to 13 weeks and 4.7 beyond. It had not levelled off in four months, so a lab's own history stops being a valid comparison within weeks.

One frame. The same measurement that places a lab can move it. Each batch's untreated wells are mapped onto the field's mean and covariance, built without that lab, in 200 principal components, and the same map is applied to the treated wells. On the positive controls, agreement with every other lab rises from 0.75 to 0.78, in 121 of 125 batches; the margin over the wrong compound rises in 125 of 125, so the map does not make every compound look alike. Normalised to the batch's own controls, the usual practice, agreement is 0.76, and the frame is above that in 107 of 125 batches. The gain is between labs: agreement with a lab's own first batch is unchanged (0.88, 0.88) and the spread of effect size between batches of a lab is 0.06 as measured, 0.20 on own controls and 0.12 in the frame (results/e12).

Density. Two references built from one lab each sit 6.2 within-plate spreads apart; from 5 labs each, 2.8, close to one over root n. A lab's own baseline settles by eight plates (0.89 to 0.39). The reference is what tightens with every lab that joins (results/e13).

Replay. Every batch of every lab was replayed as an upload and judged against the published field, each lab scaled on the other ten. A plate is outside when it sits further from its lab's own first batch than from the nearest other lab's centre. 6 of 129 batches are outside, in 3 of 11 labs. In lab 2 the first is 30 August 2021, 5 weeks before the October batch; its cell count is inside the lab's own range. Lab 2's verdicts are unchanged under nine of ten variants of the analysis (results/e9).

What moved. The distance is a difference of two means, so it reads feature by feature. On 30 August: ER and AGP stain intensity up 2.9 and 2.8 standard deviations, Mito up 1.4. Cell-level features carry 80% of the shift. Profiles cannot tell a stain lot from biology; they say where to look (results/e6).

Does it matter. JUMP runs the same positive controls on every compound plate. Inside labs, the further a batch has drifted from its lab's baseline, the less its measured drug effects agree with that baseline: Spearman −0.71, 118 batches, 11 labs, 1,829 plates; −0.03 when batches are shuffled inside their lab; −0.72 when drift and effect use different wells (results/e10).

The known answer. In lab 3, 4 consecutive batches, 65 plates, the wells the public plate map labels as positive controls show no effect, and no other well on those plates does. Untreated wells flagged only 1 of those 4 batches, which is why a certificate has two halves (results/e11).

Why correction is not the answer. Batch correction for image-based profiling has been benchmarked on JUMP across ten methods [4]; its purpose is to remove lab and batch differences so biology can be compared. That is right for discovery. For quality control it deletes the one coordinate that says where a lab sits.

4 Formalism

Everything in section 3 is one of six operations on the same well vector, each a difference of means or a held-out classifier, so any lab can check it. Every well becomes an embedding, z-scored across the field:

$$e = \varphi(image) \in \mathbb{R}^{3773} \qquad \varphi \in \{CellProfiler,\ CNN,\ ViT\} \tag{2}$$

OPERATION	DEFINITION	ROLE
1 · Embedding	$e = \varphi(image)$	the sensor; choice of φ barely matters [6]
2 · Lab classifier	$f : e \to lab$	does the lab leave a fingerprint (99.9%)
3 · Certificate distance	eq. (3)	how far a plate sits from the field, and which way

OPERATION	DEFINITION	ROLE
4 · Variance decomposition	$Var(e) = V_{model} + V_{site} + V_{batch} + V_{plate} + V_{well}$	how much is instrument, site, batch
5 · Residualize known variables	$e_{\perp} = e - \hat{E}[e \mid c]$	remove only what is known to be instrument
6 · Morphology change	$\delta = f(\lVert e - E[e \mid vehicle] \rVert)$	size of a shift, for the known-answer test

The certificate is a difference of two means in units of the field's spread, conditioned on the recorded variables c (cell line, plate type, microscope, and so on):

$$d = \lVert E[e \mid plate] - E[e \mid field, c] \rVert_{\Sigma} \quad (3)$$

Its direction names the features that moved; its negative points back toward the field. "Normal" is therefore a conditional expectation, $E[x \mid every\ variable]$, and it is only as good as the variables recorded. JUMP records the instrument side (site, batch, plate type, microscope, objective, filter set, pixel size); it does not publicly record serum or media lot, passage, seeding density, dye lots or operator.

Discovery and QC want opposite things from the same vector. Discovery wants invariance to the lab, so it aligns every plate to a reference before comparing. QC wants the lab coordinate, so it must never align it away.

5 Limits and open tests

Each claim has a named test that could fail. Two have now been run.

OPEN QUESTION	TEST	WHAT WOULD FALSIFY IT	STATUS
Does label-free brightfield carry the lab signal?	Re-run e1 on the brightfield planes of the same 93,228 wells	Brightfield near 9% chance: the certificate needs one stain	pending

OPEN QUESTION	TEST	WHAT WOULD FALSIFY IT	STATUS
Does a lab's position change the answer for a fixed drug?	e8: 8 positive controls on 1,829 plates, 11 sources; effect vs the field's, against control distance	Spearman near zero: cells differ but answers do not	run: −0.71 inside labs; across 11 labs the sign agrees and the count is too small
Is the fingerprint biology or instrument?	Variance decomposition, microscope model separated from lab	Nearly all variance in V_{model}: a calibration product	in part: four labs on one microscope model separate at 99.99%; decomposition pending
Does drift generalize beyond one lab?	Repeat e1b at each partner lab	Unbounded drift elsewhere: no stable reference	one lab; still rising at four months

Already controlled: vehicle-only wells (no compounds, so no treatment signal to leak), folds held out by plate, a permuted-label baseline at chance (9.3%), and the result surviving removal of the 1536-well labs.

6 Industry position

The market checks cell identity and ignores cell behaviour. Datum Line is the behaviour half, held by nobody who sells cells, media or instruments.

OFFERING TODAY	WHAT IT PROVES	WHAT IT CANNOT SEE	RELATION
STR authentication	which line it is	same fingerprint, different behaviour	complement: identity, then behaviour
Vendor certificate of analysis	the lot shipped to spec	your incubator, after passage	complement
Instrument software, plate controls	the plate against itself	a whole plate, batch or site off	reads the same images

OFFERING TODAY	WHAT IT PROVES	WHAT IT CANNOT SEE	RELATION
Phenomics platforms, batch correction	biology, lab removed on purpose	the lab coordinate, by design	collaborator: discovery upstream
Eyeballing and cell counts	gross death or overgrowth	texture and shape at equal counts	what labs do now

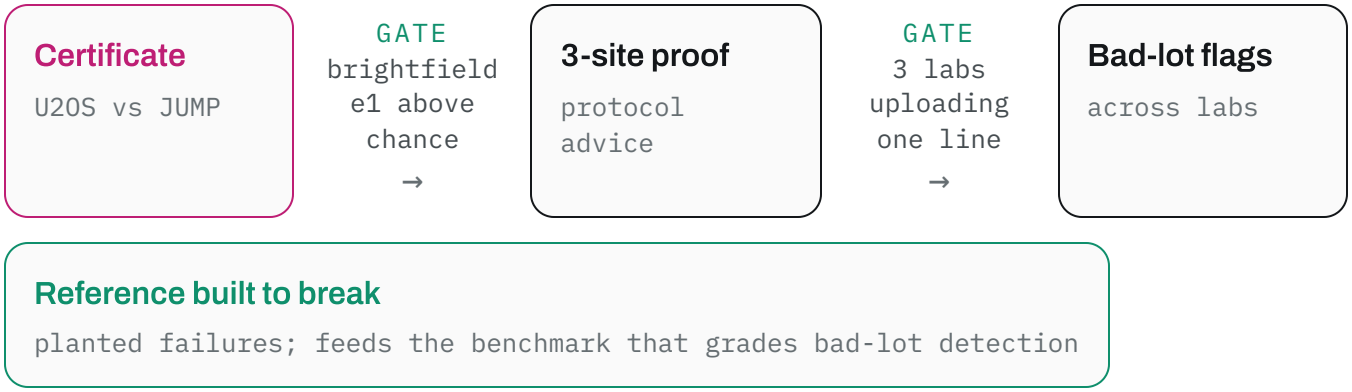
Who pays. Academic labs and cores, on every new line (at stake: the paper). Pharma and CRO screening groups, on every transfer (at stake: the screen; one line checked late cost about $2.6M in the author's experience: 3 runs a week for 17 weeks at about $50k a run). Cell banks and vendors, on every lot (at stake: the lot and every lab that bought it).

Why it compounds. Day one needs no network: the public JUMP reference already holds eleven labs' U2OS controls, so the first customer is compared with the field, not with an empty database. Every upload widens the reference, and bad-lot detection needs many labs on the same lot, so the later products exist only once the network does.

Price. Uploading is free: a lab puts its cells on the map and gets its certificate. The paying side is the company that buys cell data it did not make and needs it comparable: a fee per certificate on the data it bought, and a subscription per line, per quarter for groups that screen continuously. Figures are not yet set. **What leaves the building:** untreated control-well images only; no compounds, targets or results.

7 Roadmap

Fig. 3 Each phase opens only when the test before it passes



Gates are falsifiable tests from section 5; size and budget of the reference set are not yet set.

References

References 1 to 5 were opened and checked for this paper. Reference 6 is the author's own work. Results files (results/, produced by the scripts in analysis/) are the author's analyses of cpg0016.

1. Freedman LP, Cockburn IM, Simcoe TS. The economics of reproducibility in preclinical research. *PLoS Biology* 13(6):e1002165, 2015. PMC4461318
2. Bray MA, Singh S, Han H, et al., Carpenter AE. Cell Painting, a high-content image-based assay for morphological profiling using multiplexed fluorescent dyes. *Nature Protocols* 11:1757–1774, 2016. doi:10.1038/nprot.2016.105
3. Chandrasekaran SN, et al. JUMP Cell Painting dataset: morphological impact of 136,000 chemical and genetic perturbations. *bioRxiv* 2023.03.23.534023, 2023. bioRxiv
4. Arevalo J, Su E, Ewald JD, van Dijk R, Carpenter AE, Singh S. Evaluating batch correction methods for image-based cell profiling. *Nature Communications* 15:6516, 2024. doi:10.1038/s41467-024-50613-5
5. Seal S, Dee W, ... Titterton K, ... Carpenter AE. Counting cells can accurately predict small-molecule bioactivity benchmarks. *Nature Communications*, 2026. doi:10.1038/s41467-026-68725-5
6. Ewald JD, Titterton KL, et al. Cell Painting for cytotoxicity and mode-of-action analysis in primary human hepatocytes. *Cell Systems*, 2026.